8.02 Number Theory			
8.02a	Number bases	a) Understand and be able to work with numbers written in base n, where n is a positive integer. *The standard notation for number bases will be used.* *i.e.* 2013_n *will denote the number* $2n^3 + n + 3$ *(with* $n > 3$ *in this example) and the letters* A*-*F *will be used to represent the integers 10-15 respectively when* $11 \leq n \leq 16$*.*	
8.02b	Divisibility tests	b) Be able to use (without proof) the standard tests for divisibility by $2, 3, 4, 5, 8, 9$ and 11. *Includes knowing that repeated tests can be used to establish divisibility by composite numbers.*	
8.02c		c) Be able to establish suitable (algorithmic) tests for divisibility by other primes less than 50. *For integers* a *and* b*, the notation* $a \mid b$ will be used for *"*a *divides exactly into* b*" ("*a *is a factor of* b*", "*b *is a multiple of* a*", etc.).*	
8.02d	The division algorithm	d) Appreciate that, for any pair of positive integers a, b with $0 < b \leq a$, we can uniquely express a as $a = bq + r$ where q (the quotient) and r (the residue, or remainder, when a is divided by b) are both positive integers and $r < b$.	

OCR Ref.	Subject Content	Stage 1 learners should...	Stage 2 learners should additionally...
8.02e 8.02g 8.02f 8.02h	Finite (modular) arithmetics	e) Understand and be able to use finite arithmetics (the arithmetic of integers modulo n for $n \geq 2$). f) Be able to solve single linear congruences of the form $ax \equiv b \pmod{n}$.	g) Be able to calculate quadratic residues and solve, or prove insoluble, equations involving them. h) Be able to solve simultaneous linear congruences of the form $ax \equiv b \pmod{n}$. *No more than three simultaneous linear congruences will be used. Use of the Chinese remainder theorem will be allowed but not required.*
8.02i 8.02j	Prime numbers	i) Understand the concepts of prime numbers, composite numbers, highest common factors (hcf), and coprimality (relative primeness). *Knowledge of the fundamental theorem of arithmetic will be expected, but proof of the result will not be required.* j) Know and be able to apply the result that $a \mid b$ and $a \mid c \Rightarrow a \mid (bx + cy)$ for any integers x and y. *Includes using this result, for example to test for common factors or coprimality.*	
8.02k	Euclid's lemma	k) Know and be able to use Euclid's lemma: if $a \mid rs$ and $\text{hcf}(a, r) = 1$ then $a \mid s$.	

OCR Ref.	Subject Content	Stage 1 learners should…	Stage 2 learners should additionally…
8.02l	Fermat's little theorem		l) Know and be able to use Fermat's little th forms: 1. if p is prime and hcf $(a, p) = 1$, then $a^{p-1} \equiv$ 2. if p is prime $a^p \equiv a\,(\text{mod } p)$. *Includes recognising that if p is prime then Ferm theorem holds, but that the converse is not true aware of the existence of "pseudo-primes" to ba* [*The proof is excluded.*] [*Carmichael numbers are excluded.*]
8.02m 8.02n	The order of *a* modulo p		m) Know and be able to use the fact that $p -$ necessarily the least positive integer n for $a^n \equiv 1(\text{mod } p)$. n) Know that the n with this property, called a modulo p", is a factor of $p-1$ and be ab specific cases.
8.02o	Binomial theorem		o) Be able to use the binomial theorem to sh $(a+b)^p \equiv a^p + b^p(\text{mod } p)$ for prime p, an result.